Helix Genomic Reference Laboratory

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CNS TUMOR PROFILE NGS PANEL (172-GENE) — FINAL REPORT

PATIENT	SPECIMEN	REQUESTED BY
MARTINEZ-LANG, ELENA J. MRN MRN-3318477 | DOB 1979-06-02 47 y / Female	Reference accession HX-NGS-CNS-018244 Client accession SP-26-N-04412 FFPE block A4 (tumor) Collected 2026-04-04	Northshore University Hospital D. Sokolov, MD Received 2026-04-11 Reported 2026-04-22

CLINICAL HISTORY (PROVIDED BY ORDERING SITE)

47-year-old woman with six-week history of new-onset focal motor seizures and progressive right-hand clumsiness. MRI brain demonstrated a 3.8 cm right frontal lobe mass with patchy enhancement and moderate vasogenic edema. No prior cancer history, no prior CNS irradiation. Family history negative for CNS tumors. Underwent right frontal craniotomy with gross total resection on 2026-04-04.

RESULT SUMMARY

POSITIVE — IDH1 R132H (Tier I, classifying), **ATRX** truncating (Tier I), **TP53 R175H** (Tier II, non-classifying), **MGMT promoter unmethylated**, **CDKN2A/B** no homozygous deletion. See variant table and interpretation.

DETECTED VARIANTS

Gene	Variant (HGVS)	Consequence	VAF	Tier
IDH1	`NM_005896.4:c.395G>A` `p.(Arg132His)`	Missense, R132 hotspot; defining for IDH-mutant astrocytoma	**42%**	**I**
ATRX	`NM_000489.6:c.6018del` `p.(Lys2007AsnfsTer4)`	Frameshift / truncating; concordant with IHC loss	**41%**	**I**
TP53	`NM_000546.6:c.524G>A` `p.(Arg175His)`	Missense, R175 hotspot; co-occurring; non-classifying	**38%**	**II**

Tier I = strong clinical / classifying significance. Tier II = potential clinical significance (prognostic / co-occurring). VAF = variant allele frequency.

COPY-NUMBER ANALYSIS (FROM PANEL DATA)

Locus	Copy state	Interpretation
CDKN2A/B (9p21.3)	Diploid (2 copies)	NO homozygous deletion. Not a grade-4 driver in this case.
EGFR (7p11.2)	Diploid	No amplification (would suggest IDH-wt GBM).
Chromosome 7	Diploid	No +7 (would suggest IDH-wt GBM).
Chromosome 10	Diploid	No -10 (would suggest IDH-wt GBM).

MGMT PROMOTER METHYLATION

Method	Result
Pyrosequencing (4 CpGs averaged)	**UNMETHYLATED** (mean 4.1%; cutoff <10%)

MGMT promoter methylation is a predictive biomarker for response to alkylating chemotherapy (temozolomide). An unmethylated result is associated with reduced anticipated benefit. This finding does not change the disease category.

GENES WITHOUT REPORTABLE VARIANT

No reportable variant detected in: H3F3A (H3 G34, H3 K27), HIST1H3B, TERT promoter, BRAF (V600 and fusions), FGFR1/2/3, NF1, NF2, PIK3CA, PIK3R1, PTEN, RB1, MYC, MYCN, MET, CIC, FUBP1, NOTCH1, SMARCB1, and the remaining 152 genes on panel. (Full coverage report retained by the laboratory.)

INTERPRETATION

This CNS tumor NGS panel identifies a **canonical IDH1 R132H missense mutation** and a **truncating ATRX variant** (concordant with the IHC loss reported on the morphology specimen), together with a **TP53 R175H missense variant**. The constellation of IDH mutation + ATRX loss + TP53 mutation in a diffuse astrocytic glioma supports an **astrocytoma, IDH-mutant** entity per WHO CNS5 (2021). 1p/19q integrity (assessed by neuropathology FISH on the same specimen) excludes oligodendroglioma. Importantly, **CDKN2A/B is diploid** — there is no homozygous deletion. Under WHO CNS5 grading rules, this absent grade-4 molecular criterion supports a grade of 2 or 3 (final grading depends on histology, specifically the presence of mitotic activity, which is described on the morphology report). No EGFR amplification, no +7/-10 co-occurrence, and no TERT promoter mutation: not consistent with IDH-wildtype glioblastoma. MGMT promoter is **unmethylated**, informing therapy planning. TP53 R175H is a hotspot pathogenic variant and is reported here for prognostic completeness; it is <u>not</u> a classifying variant in WHO CNS5 and should not, by itself, change the entity name in the integrated diagnosis.

METHODOLOGY & LIMITATIONS

Hybrid-capture targeted NGS of 172 CNS-tumor-associated genes performed on tumor DNA extracted from FFPE block (cassette A4). Mean target coverage 980x; minimum 250x. Analytical sensitivity approximately 5% VAF for SNVs and small indels. Copy-number analysis derived from per-gene normalized read depth; reportable for high-confidence gains/losses. The assay does not reliably detect structural rearrangements outside targeted breakpoints, balanced translocations, or epigenetic events apart from MGMT (reported by orthogonal pyrosequencing). A negative result does not exclude a low-level variant or a variant in a region of low coverage. Helix Genomic Reference Laboratory validated this laboratory-developed test; it has not been cleared or approved by the FDA.

Electronically signed: Ranganathan, Vimal S., MD, PhD, Laboratory Director, and Park, R., PhD, Variant Scientist — 2026-04-22. Results released to ordering institution for integration into the combined diagnostic report.